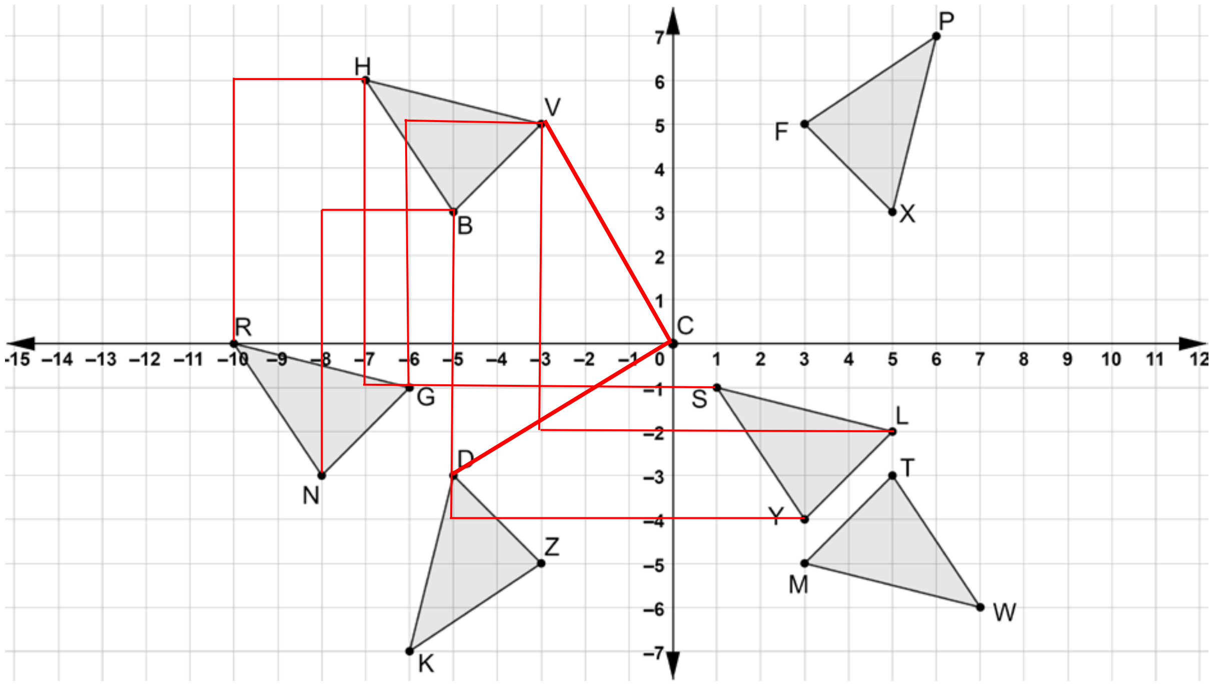
Geometry
Unit 4
Lesson 1 Practice

Name_Answer Key________
Date____________________
Period__________________

Six triangles are shown on the coordinate grid.



1. Draw the mapping that shows the translation if $\triangle SYL$ is the preimage and $\triangle HBV$ is the image.

2. Draw the mapping that shows the translation if $\triangle HBV$ is the preimage and $\triangle RNG$ is the image.

3. Describe the translation if $\triangle HBV$ is the preimage and $\triangle RNG$ is the image.
$\triangle HBV$ is shifted left 3 units and down 6 units.

4. Complete the statements.

The x-values of $\triangle RNG$, the image, can be found by

- o adding
- o subtracting

__11__ unit(s) to the x-values of $\triangle SYL$, the preimage. The y-values of $\triangle RNG$, the image, can be found by

- o adding
- o subtracting

__1__ unit(s) to the y-values of $\triangle SYL$, the preimage.

5. On the coordinate grid draw angle DCV.

6. The algebraic description of the rotation of $\triangle KZD$, the preimage, onto $\triangle HBV$ is the image, is $(x, y) \rightarrow (y, -x)$. Describe the rotation. Then, use the coordinates of the vertices to show how the rotation matches the algebraic description.

90° clockwise rotation	K: $(-6, -7)$	H: $(-7, 6)$
	Z: $(-3, -5)$	B: $(-5, 3)$
	D: $(-5, -3)$	V: $(-3, 5)$

7. The algebraic description of the rotation of $\triangle TWM$, the preimage, onto $\triangle BHV$ is the image, is $(x, y) \rightarrow (-x, -y)$. Describe the rotation. Then, use the coordinates of the vertices to show how the rotation matches the algebraic description.

180° rotation	T: $(5, -3)$	B: $(-5, 3)$
	W: $(7, -6)$	H: $(-7, 6)$
	M: $(3, -5)$	V: $(-3, 5)$

8. Describe how the measure of $\angle BCF$ can be used to determine the degrees of rotation where $\triangle PFX$ is the preimage and $\triangle HBV$ is the image.

$\angle BCF$ is a 90° angle. It shows that $\triangle PFX$ is rotated 90° counterclockwise.

9. Describe how the coordinates of the vertices of $\triangle PFX$, the preimage, and the coordinates of $\triangle HBV$, the image can be used to determine the degrees of rotation.

The y-coordinate is multiplied by -1, then the x and y-values switch.

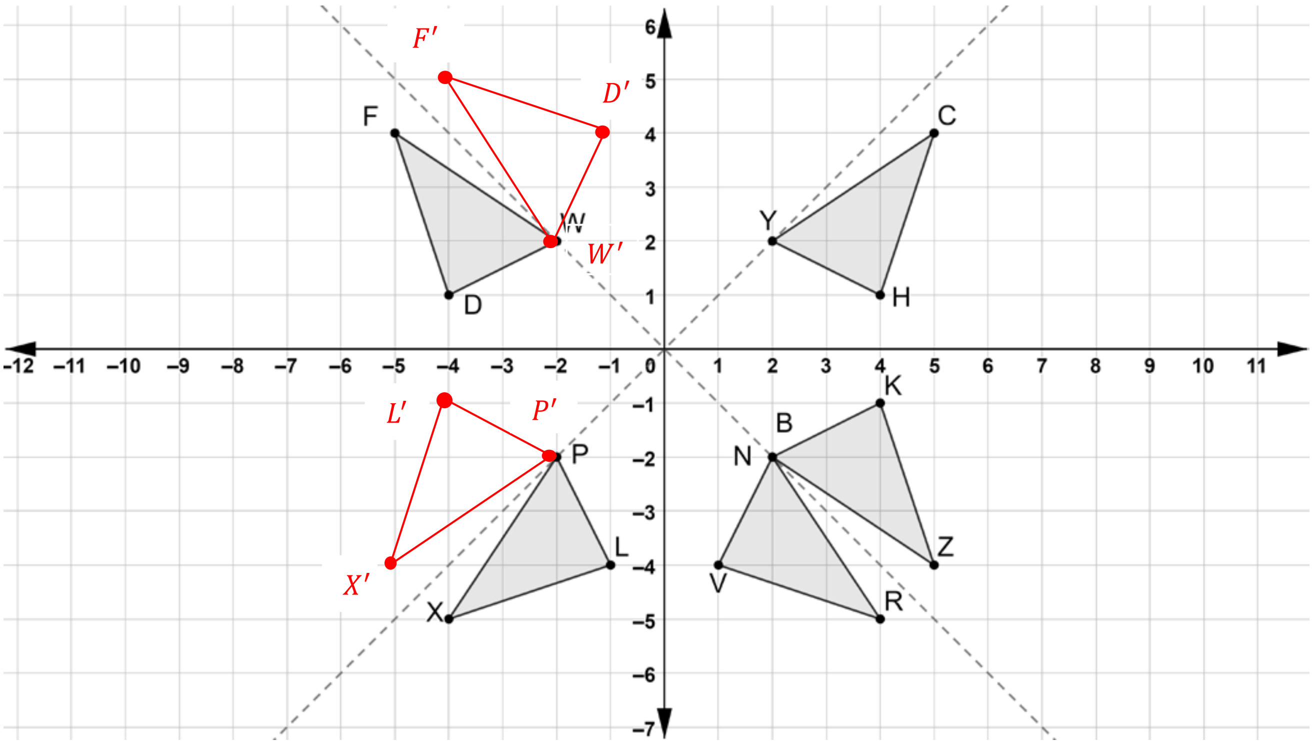
10. Describe how the orientation of the triangles helps to determine which transformations were a translation and which transformations were a rotation.

Translations do not change the orientation of a figure, while rotations do change the orientation.

Geometry
Unit 4
Lesson 2 Practice

Name_Answer Key________
Date___________________
Period__________________

Six triangles are shown on the coordinate grid.



1. Draw the reflection of $\triangle PLX$ across the line $y = x$. Label the reflection as $\triangle P'L'X'$.

2. Draw the reflection of $\triangle FWD$ across the line $y = -x$. Label the reflection as $\triangle F'W'D'$.

3. Select all of the triangles that are reflections of $\triangle CHY$.
   - ☐ $\boxed{\triangle FDW}$
   - ☐ $\boxed{\triangle XLP}$
   - ☐ $\triangle RVN$
   - ☐ $\boxed{\triangle ZKB}$
   - ☐ $\triangle P'L'X'$
   - ☐ $\triangle F'W'D'$

4. The algebraic description of the reflection of $\triangle BKZ$, the preimage, onto $\triangle NVR$, the image, is $(x, y) \rightarrow (-y, -x)$. Describe how the coordinates of the vertices of $\triangle BKZ$, the preimage, and the coordinates of $\triangle NVR$ the image can be used to determine the line of reflection.

When the coordinates switch places and become opposite, the figure is reflected over the line $y = -x$.

5. Which triangle is the image of reflecting $\triangle FDW$ across the x-axis?

$\triangle X'L'P'$

6. Which triangle is the image of reflecting $\triangle CHY$ across the y-axis?

$\triangle FDW$

7. Describe how the coordinates of the vertices of $\triangle F'W'D'$, the preimage, and the coordinates of $\triangle XPL$ the image can be used to determine the algebraic description of the reflection.

The x-coordinates of $\triangle F'W'D'$ do not change and the y-coordinates become opposite creating the algebraic description $(x, y) \rightarrow (x, -y)$.

8. Describe how the coordinates of the vertices of $\triangle ZKB,$ the preimage, and the coordinates of $\triangle CHY$, the image can be used to determine the algebraic description of the reflection.

The x-coordinates of $\triangle ZKB$ do not change and the y-coordinates become opposite creating the algebraic description $(x, y) \rightarrow (x, -y)$.

9. Write the algebraic description for the reflection of $\triangle NVR$, the preimage, onto $\triangle PLX$, the image.

$(x, y) \rightarrow (-x, y)$

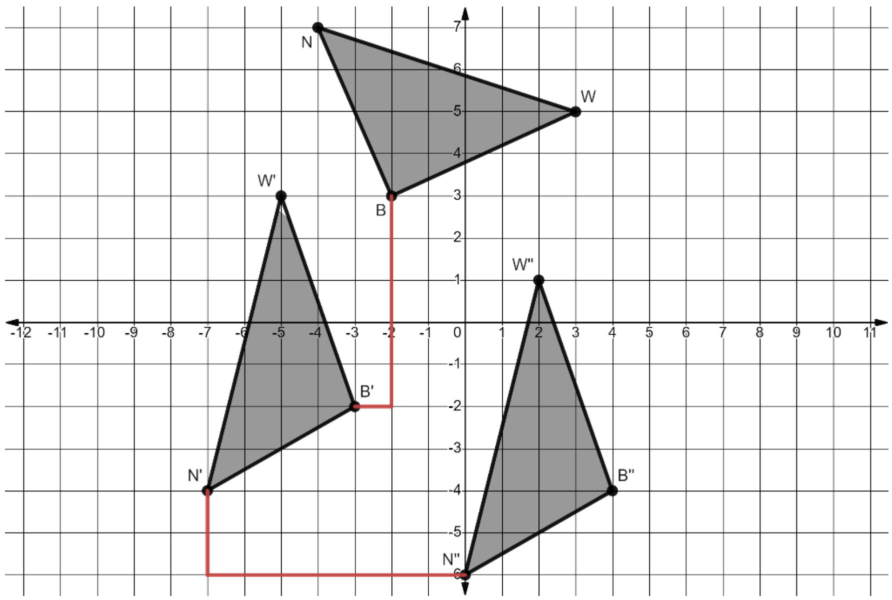
10. Write the algebraic description for the reflection of $\triangle P'L'X'$, the preimage, onto, $\triangle BKZ$, the image.

$(x, y) \rightarrow (-x, y)$

Geometry
Unit 4
Lesson 3 Practice

Name_Answer Key________
Date____________________
Period__________________

Triangles BWN, $B'W'N'$, and $B''W''N''$ are shown on the coordinate grid where $\triangle BWN$ is the preimage.



1. Describe the feature of $\triangle B'W'N'$ that indicates the rigid motion that was applied to $\triangle BWN$ that resulted in $\triangle B'W'N'$.
   Rotation $90°$ counterclockwise about the origin.

2. Show the feature described by mapping the transformation of one vertex of $\triangle BWN$ to $\triangle B'W'N'$.

3. Complete the table.

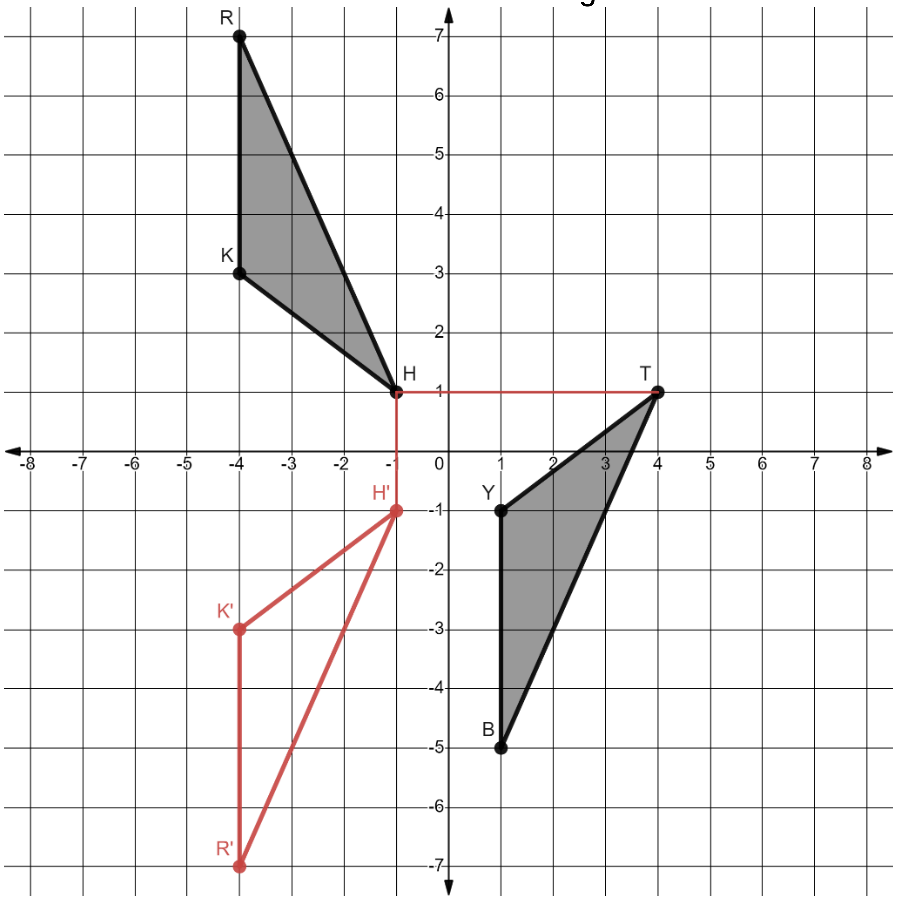
Transformation	Mapping	Algebraic Description
Rotation $90°$ counterclockwise about the origin.	$\triangle BWN \rightarrow \triangle B'W'N'$	$(x, y) \rightarrow (-y, x)$

4. Map the transformation of one vertex of $\triangle B'W'N'$ to $\triangle B''W''N''$.

5. Complete the table.

Transformation	Mapping	Algebraic Description
Translation right 7, down 2	$\triangle B'W'N' \rightarrow \triangle B''W''N''$	$(x, y) \rightarrow (x + 7, y - 2)$

Triangles RKH and BYT are shown on the coordinate grid where $\triangle RKH$ is the preimage.



6. Describe the feature of $\triangle TYB$ that indicates that more than one rigid motion was applied to $\triangle RKH$.
   The orientation and location changed.

7. Draw $\triangle R'K'H'$, the result of the first transformation that was applied to $\triangle RKH$.

8. Map the transformation of one vertex of $\triangle R'K'H'$ to $\triangle BYT$.

9. Complete the table.

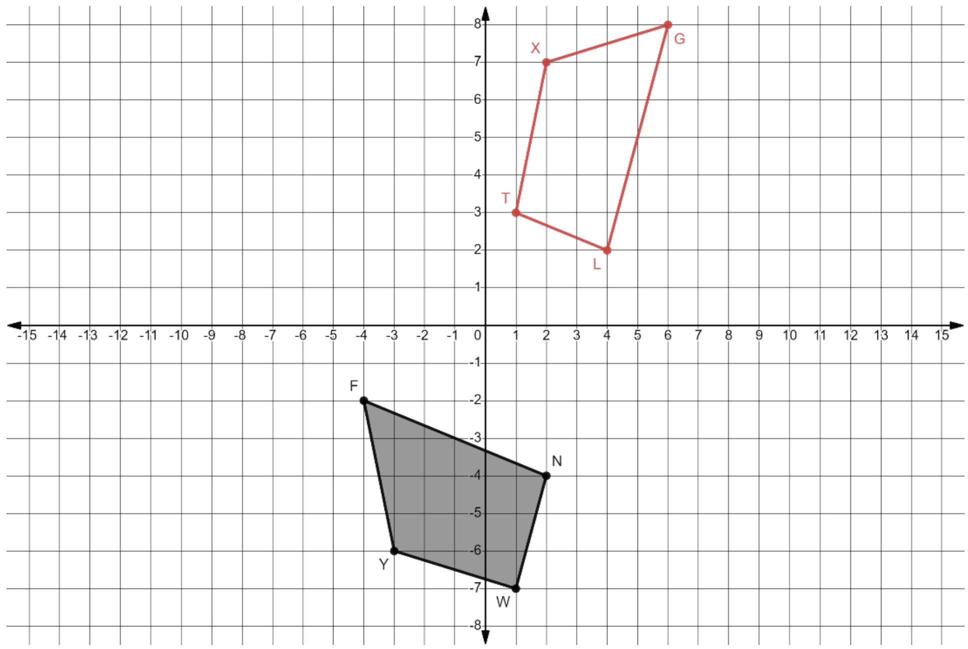
Transformation	**Mapping**	**Algebraic Description**
Reflect over x-axis	$\triangle RKH \to \triangle R'K'H'$	$(x, y) \to (x, -y)$
Translate right 5 up 2	$\triangle R'K'H' \to \triangle BYT$	$(x, y) \to (x + 5, y + 2)$

10. Explain how to know if the sequence of transformations preserves distance.
    The distance between each vertex in the image is the same.

Geometry
Unit 4
Lesson 4 Practice

Name_Answer Key________
Date___________________
Period__________________

Quadrilateral $FNWY$ is shown on the coordinate grid.



1. Draw the quadrilateral that is the result of a translation left 4 units and up 8 units and then rotated $90°$ clockwise about the origin. Label the new quadrilateral $GLTX$.

Complete the table.

	Transformation	Mapping	Algebraic Description
2.	Translation left 4, up 8	$FNWY \rightarrow F'N'W'Y'$	$(x, y) \rightarrow (x - 4, y + 8)$
3.	Rotation $90°$ clockwise	$F'N'W'Y' \rightarrow GLTX$	$(x, y) \rightarrow (y, -x)$

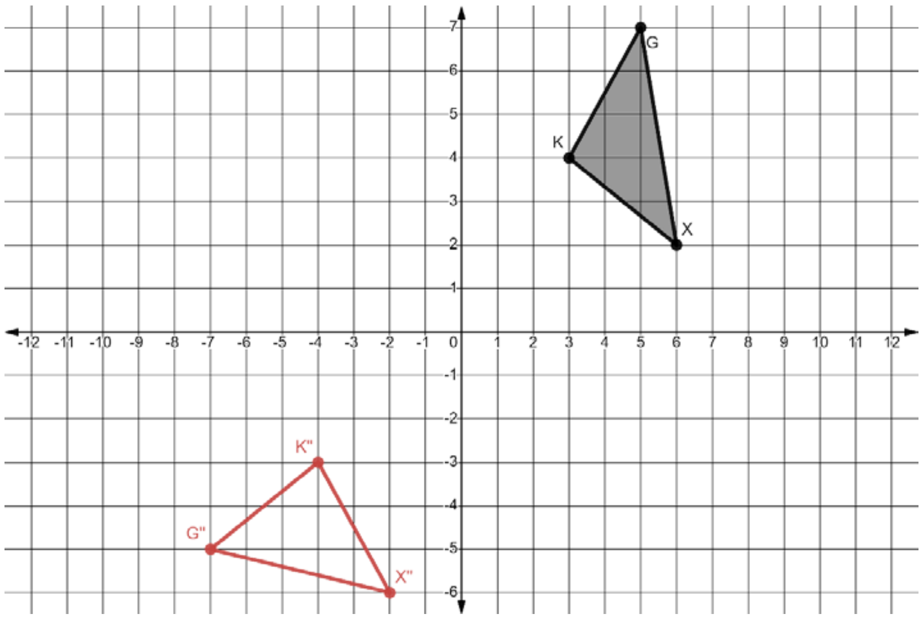
4. Complete the table of values for each quadrilateral.

Quadrilateral $FNWY$			Quadrilateral $F'N'W'Y'$			Quadrilateral $GLTX$		
Vertex	x	y	**Vertex**	x	y	**Vertex**	x	y
F	-4	-2	F'	-8	6	G	6	8
N	2	-4	N'	-2	4	L	4	2
W	1	-7	W'	-3	1	T	1	3
Y	-3	-6	Y'	-7	2	X	2	7

5. Explain how the tables for quadrilaterals $FNWY$ and $F'N'W'Y'$ show the effect of the transformation on the coordinates that result in the algebraic description.
The translation subtracts 4 from the x-coordinate and adds 8 to the y-coordinate.

6. Explain how the tables for quadrilaterals $F'N'W'Y'$ and $GLTX$ show the effect of the transformation on the coordinates that result in the algebraic description.
The rotation changes the x-coordinate to its opposite, then switches the coordinates.

Triangle XGK is shown on the coordinate grid.



7. Draw the triangle that is the result of a rotation $270°$ counterclockwise about the origin and then a reflection across the y-axis. Label the new triangle $X''G''K''$.

Complete the table.

	Transformation	Mapping	Algebraic Description
8.	Rotation 270° counterclockwise	$\triangle XGK \rightarrow \triangle X'G'K'$	$(x, y) \rightarrow (y, -x)$
9.	Reflection over y-axis	$\triangle X'G'K' \rightarrow \triangle X''G''K''$	$(x, y) \rightarrow (-x, y)$

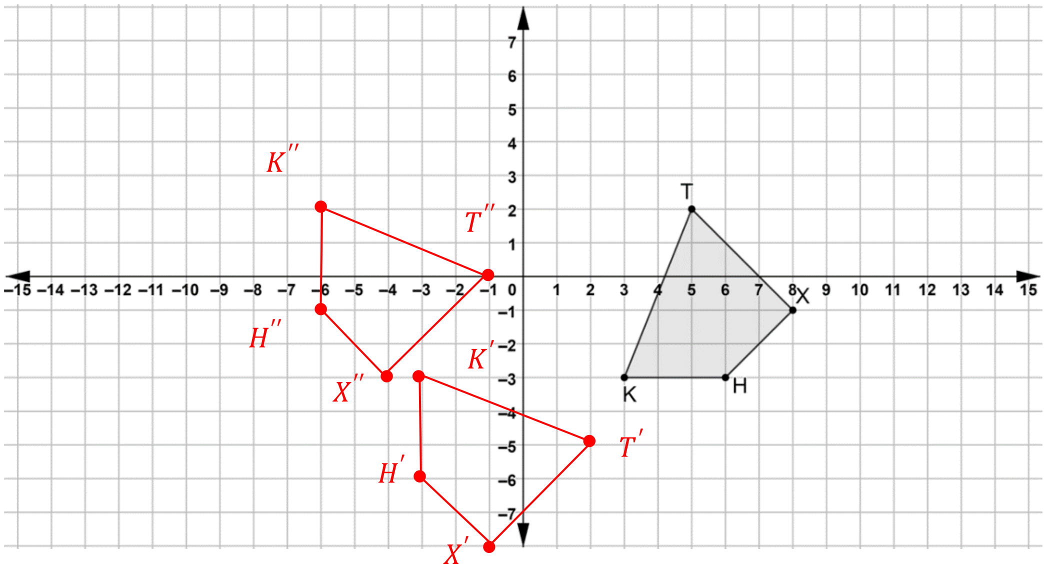
10. Describe the connection between the algebraic description of the rotation and the vertices of $\triangle XGK$ and $\triangle X'G'K'$.
The rotation changes the x-coordinate to its opposite, then switches the coordinates.

Geometry
Unit 4
Lesson 5 Practice

Name_Answer Key________
Date___________________
Period__________________

Quadrilateral $TXHK$ is shown on the coordinate grid.

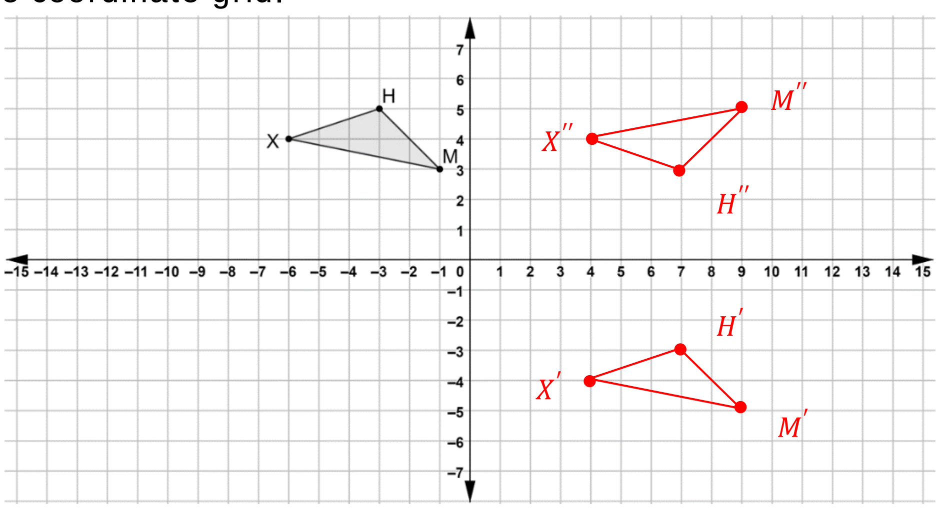
Perform the series of transformations on quadrilateral $TXHK$. Assume all rotations are centered at the origin. Label each result.



	Transformation	Quadrilateral
1.	$(x, y) \rightarrow (y, -x)$	$T'X'H'K'$
2.	$(x, y) \rightarrow (x - 3, y + 5)$	$T''X''H''K''$

Triangle XHM is shown on the coordinate grid.

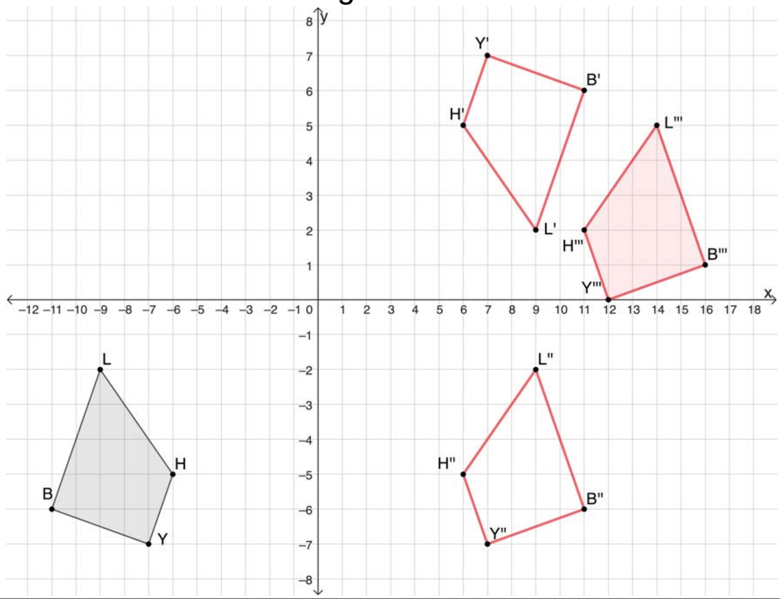
Perform the series of transformations on $\triangle XHM$. Assume all rotations are centered at the origin. Label each result.



	Transformation	Triangle
3.	$(x, y) \rightarrow (x + 10, y - 8)$	$X'H'M'$
4.	$(x, y) \rightarrow (x, -y)$	$X''H''M''$

Quadrilateral $LHYB$ is shown on the coordinate grid.

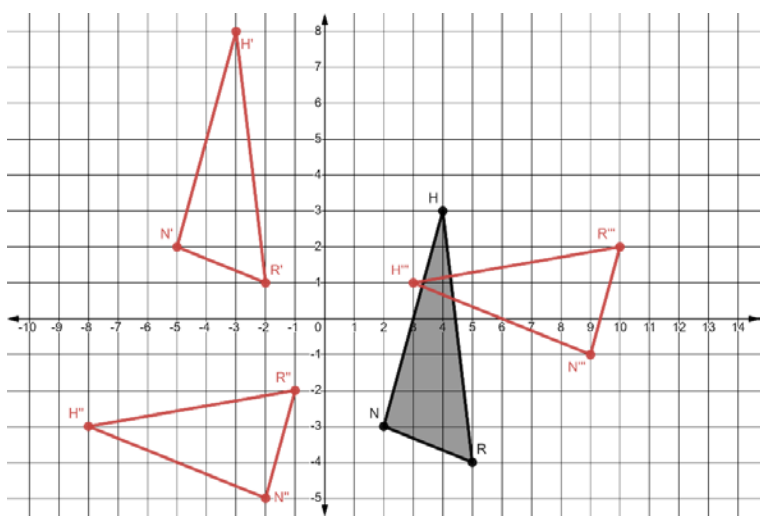
Perform the series of transformations on quadrilateral $LHYB$. Assume all rotations are centered at the origin. Label each result.



	Transformation	Quadrilateral
5.	$(x, y) \rightarrow (-x, -y)$	$L'H'Y'B'$
6.	$(x, y) \rightarrow (x, -y)$	$L''H''Y''B''$
7.	$(x, y) \rightarrow (x + 5, y + 7)$	$L'''H'''Y'''B'''$

Triangle NHR is shown on the coordinate grid.

Perform the series of transformations on $\triangle NHR$. Assume all rotations are centered at the origin. Label each result.

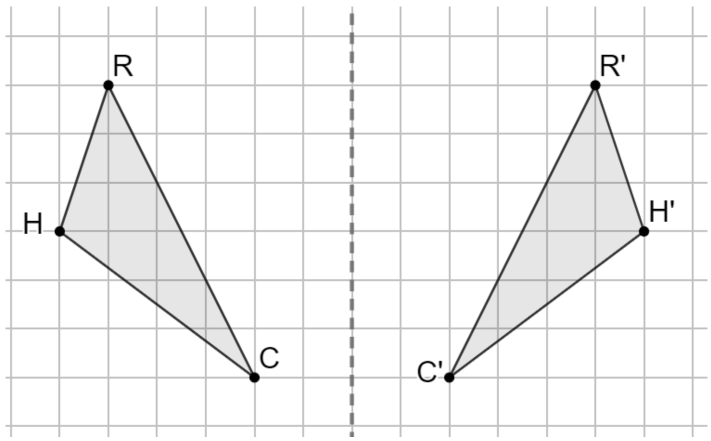


	Transformation	Triangle
8.	$(x, y) \rightarrow (x - 7, y + 5)$	$N'H'R'$
9.	$(x, y) \rightarrow (-y, x)$	$N''H'R''$
10.	$(x, y) \rightarrow (x + 11, y + 4)$	$N'''H'''R'''$

Geometry
Unit 4
Lesson 6 Practice

Name_Answer Key________
Date___________________
Period__________________

Triangles RHC and $R'H'C'$ are shown on grid paper.

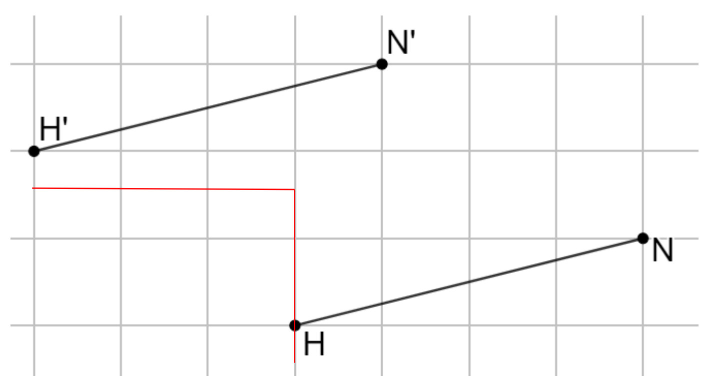


1. Explain what method(s) can be used to determine if each angle of $\triangle RHC$ has the same measure as $\triangle R'H'C'$.
Patty paper can be folded over the line of reflection to map each angle of $\triangle RHC$ to $\triangle R'H'C'$.

2. Explain what method(s) can be used to determine if each side of $\triangle RHC$ has the same length as $\triangle R'H'C'$.
Patty paper can be folded over the line of reflection to map each side of $\triangle RHC$ to $\triangle R'H'C'$.

3. Explain why the transformation of $\triangle RHC$ shown on the grid can be classified as isometry.
Because both the angle measures and side lengths are preserved.

Line segments HN and $H'N'$ are shown on the grid.



4. Show the mapping of point H to H'.

5. Describe the transformation applied to $\overline{HN}$ that resulted in $\overline{H'N'}$.
$(x, y) \rightarrow (x - 3, y + 2)$

6. Complete the statement.

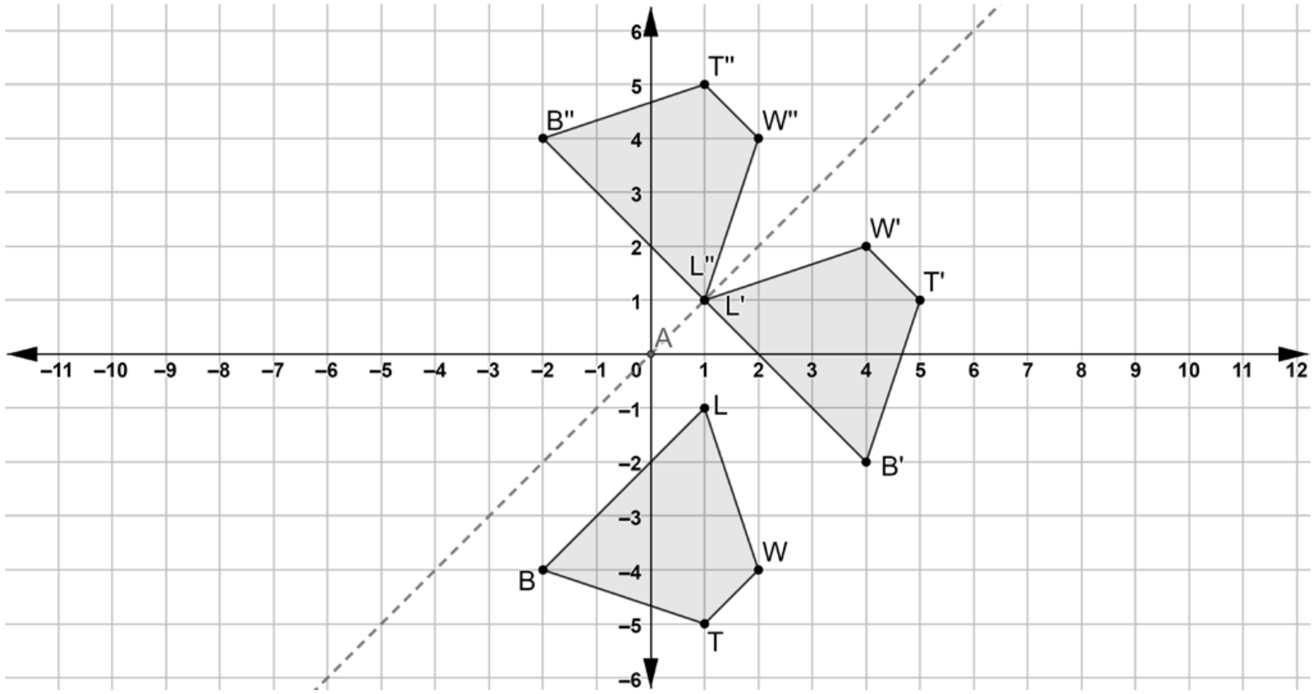
- Each point on
- Only the endpoints of

$\overline{HN}$ can be mapped to

- a corresponding point on
- the endpoints of

$\overline{H'N'}$.

Quadrilaterals $LWTB$, $L'W'T'B'$, and $L''W''T''B''$ are shown on the coordinate grid.



7. Describe what movements should be done with a patty paper copy of quadrilateral $LWTB$ so that the copy of $LWTB$ lays on top of $L'W'T'B'$.
Rotate 90° counterclockwise about the origin.

$(x, y) \rightarrow (-y, x)$

8. Describe what movements should be done with a patty paper copy of quadrilateral $L'W'T'B'$ so that the copy of $L'W'T'B'$ lays on top of $L''W''T''B''$.

Reflect over $y = x$
$(x, y) \rightarrow (y, x)$

9. Complete the statements.

The transformations used in both sequences

- o are
- o are not

examples of isometry. All of the points of

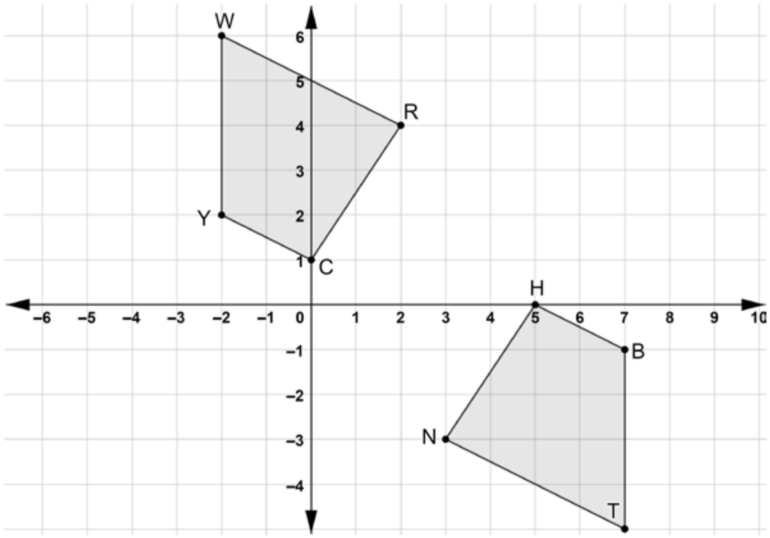
- o $L'W'T'B'$
- o $L''W''T''B''$
- o $L'W'T'B'$ and $L''W''T''B''$

can be mapped to all of the points of $LWTB$.

10. Use the definition of congruence in terms of rigid motions to explain how both sequences of transformations for quadrilateral $LWTB$ resulted in $LWTB \cong L''W''T''B''$.
Rigid motion transformations result in congruent figures.

Geometry | Name_Answer Key________
Unit 4 | Date___________________
Lesson 7 Practice | Period_________________

Quadrilaterals $CYWR$ and $BTNH$ are shown on the coordinate grid.



1. Describe a sequence of transformations that will map quadrilateral $CYWR$ onto quadrilateral $BTNH$. Rotate $180°$ clockwise about the origin, horizontal shift right 5 units, vertical shift up 1 unit.

2. The sequence of transformations that will map quadrilateral $CYWR$ onto quadrilateral $BTNH$ includes a rotation. Describe the sequence using a different rotation. Rotate $180°$ counterclockwise, horizontal shift right 5 units, vertical shift up 1 unit.

3. Complete the table that shows which vertices were mapped to each other when quadrilateral $CYWR$ was mapped to quadrilateral $BTNH$.

Quadrilateral $CYWR$	**Quadrilateral $BTNH$**
C	H
Y	B
W	T
R	N

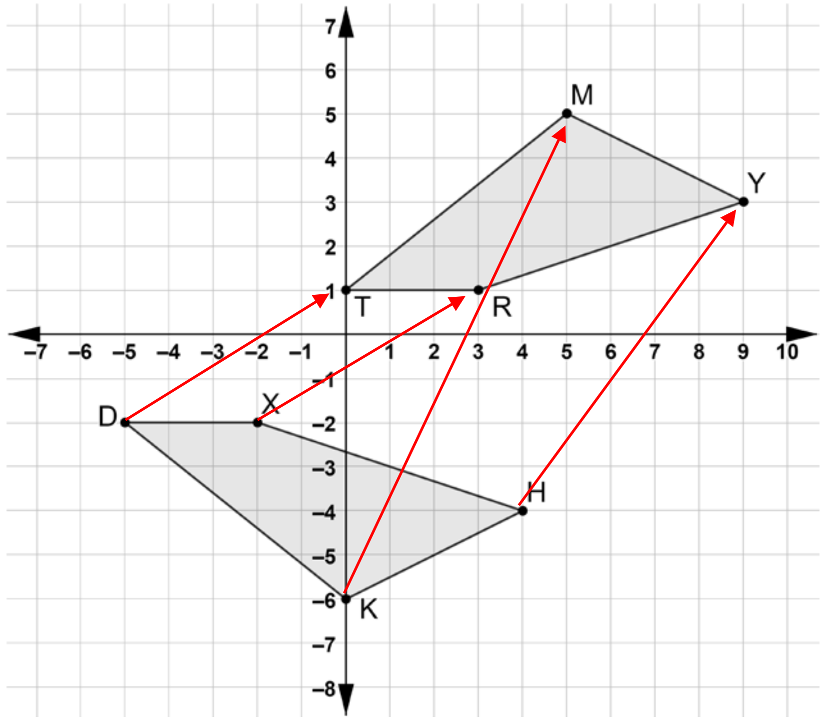
4. Write the algebraic description that matches the written description of one of the transformations.

$$(x, y) \rightarrow (-x, -y)$$

$$(x, y) \rightarrow (x + 5, y + 1)$$

5. Complete the congruence statement. $CYWR \cong HBTN$

Quadrilaterals $KHXD$ and $TMYR$ are shown on the coordinate grid.



6. Describe a sequence of transformations that will map quadrilateral $KHXD$ onto quadrilateral $TMYR$.

Reflection across x –axis, horizontal shift right 5 units, vertical shift down 1 unit.

7. Show the mapping of the transformations on the coordinate grid.

8. Complete the statement.

Vertex D maps to vertex T, vertex K maps to vertex M, vertex H maps to vertex Y, and vertex X maps to vertex R.

9. Write the algebraic description that matches the written description of the transformations.

$$(x, y) \to (x, -y)$$

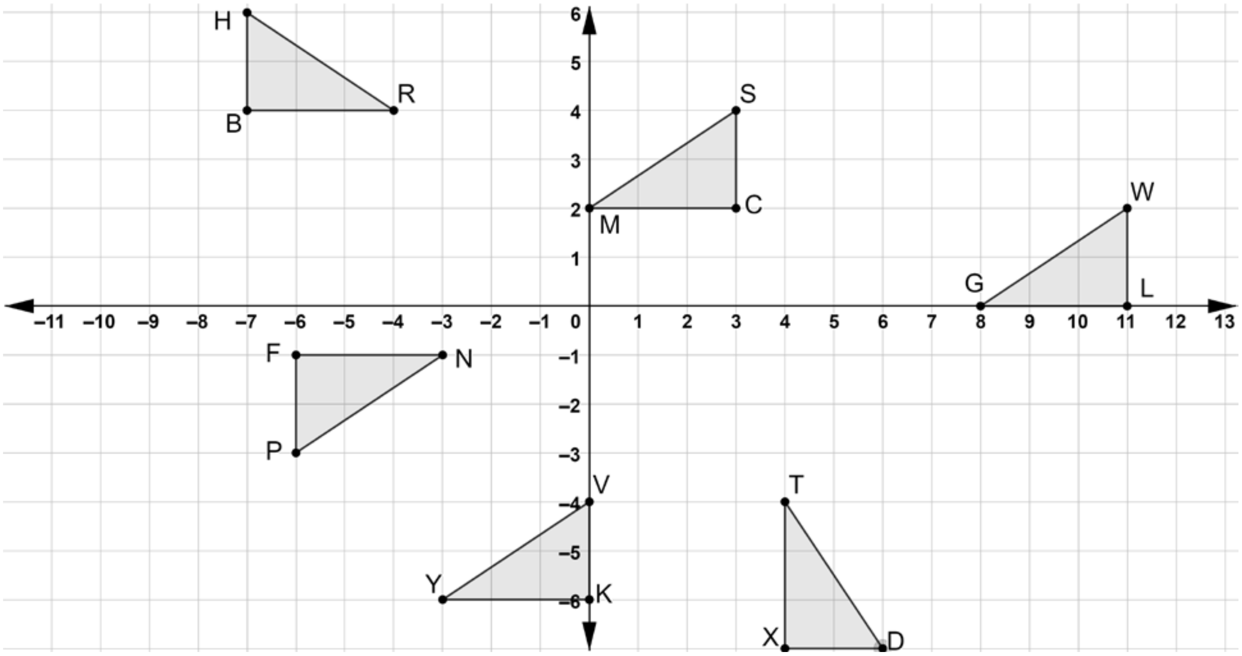
$$(x, y) \to (x + 5, y - 1)$$

10. Complete the congruence statement. $KHXD \cong MYRT$

Geometry
Unit 4
Lesson 8 Practice

Name_Answer Key________
Date___________________
Period__________________

Six triangles are shown on the coordinate grid.



Give a written description of the transformations that lead to the given congruence statement where $\triangle FNP$ is the preimage.

1. $\triangle FNP \cong \triangle CMS$

   Rotate $180°$ about the origin, translate left 3 units, and up 1 unit.

2. $\triangle FNP \cong \triangle XTD$

   Rotate $90°$ counterclockwise about the origin, translate right 3 units, and down 1 unit.

3. $\triangle FNP \cong \triangle KYV$

   Rotate $180°$ about the origin, translate left 6 units, and down 7 units.

4. $\triangle FNP \cong \triangle BRH$

   Reflect over x-axis, translate left 1 unit, and up 3 units.

5. $\triangle FNP \cong \triangle LGW$

   Rotate $180°$ about the origin, translate right 5 units, and down 1 unit.

Give an algebraic description of the transformations that lead to the given congruence statement where $\triangle FNP$ is the preimage.

6. $\triangle FNP \cong \triangle CMS$

$(x, y) \rightarrow (-x, -y)$

$(x, y) \rightarrow (x - 3, y + 1)$

7. $\triangle FNP \cong \triangle XTD$

$(x, y) \rightarrow (-y, x)$

$(x, y) \rightarrow (x + 3, y - 1)$

8. $\triangle FNP \cong \triangle KYV$

$(x, y) \rightarrow (-x, -y)$

$(x, y) \rightarrow (x - 6, y - 7)$

9. $\triangle FNP \cong \triangle BRH$

$(x , y) \rightarrow (x, -y)$

$(x, y) \rightarrow (x - 1, y + 3)$

10. $\triangle FNP \cong \triangle LGW$

$(x, y) \rightarrow (-x, -y)$

$(x, y) \rightarrow (x + 5, y - 1)$